2023 - quiz5

Graded

Student

William Chen

Total Points

10 / 10 pts

Question 1

The figure... **5** / 5 pts

✔ **– 0 pts** Correct

– 2 pts approach error

– 3 pts wrong approach

Question 2

The figure... **5** / 5 pts

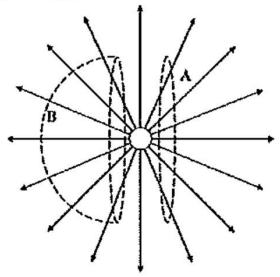
✔ **– 0 pts** Correct

Quiz 05

PH-UY 2023, Spring 2026
A Sections

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(1) (3 pts) The figure at right shows the $\vec{E}$ field lines from a single positive point charge, along with two surfaces: A = a disc of radius R, and B = a hollow-cup hemisphere also of radius R (like a hemispherical balloon with a circular opening; *not* a closed surface). The center of disc A and the center of the circular opening to hemisphere B are equidistant from the point charge. Which of those two surfaces has a greater electric flux through it (or are they equal)? Briefly justify your answer.

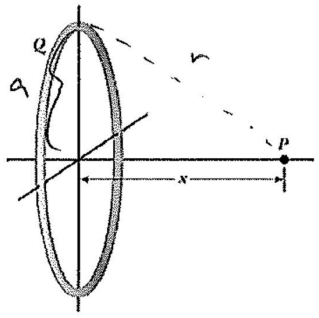


In both surfaces, the flux are EQUAL.

Flux is determined by $\oint \vec{E} \cdot d\vec{A}$, in our case, the electric field strength accross both surfaces have the same area, with the same distance from point P.

We also know that flux is determined by $\frac{\Sigma E_{enclosed}}{\varepsilon_0}$, but the surfaces are not closed surfaces, so the flux through both will be the same.

(2) (5 pts) The figure shows a uniformly charged ring of radius R and total charge Q sitting in the yz-plane (its center is at the origin). Point P is on the central axis of the ring, a distance x from the ring's center. Determine the electric field vector $\vec{E}$ at point P (give the full vector). Justify the steps you use to arrive at your answer, which should be expressed solely in terms of R, Q, x, and constants of nature, as needed.



$dE = \frac{k\,dq}{r^2}$

$E = \int dE_x = k\int \frac{dq}{r^2}\cos\theta \rightarrow \frac{x}{r} \rightarrow r = \sqrt{a^2+x^2}$

$= k\int \frac{dq}{a^2+x^2} \cdot \frac{x}{(a^2+x^2)^{\frac{1}{2}}} = \frac{kx}{(a^2+x^2)^{3/2}}\int dq = \left(\frac{kxQ}{(a^2+x^2)^{3/2}}\hat{i} = \vec{E}\right)$

To find electric field, we use our electric field equation $dE = \frac{k\,dq}{r^2}$, in our case, our vector component is the x component, so we multiply by $\cos\theta = \frac{x}{r}$. We integrate with respect to q, so take out the other variables / constants and set $\int dq = Q$ => electric charge.